

AYN THOR · LOWER-SCREEN ASSISTANT



Heimdall Installation & User Guide

AYN Thor lower-screen assistant · Quick start and feature reference

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Real-device AYN Thor use

Good to know Installing Heimdall does not require root or Developer options. Developer options are needed only for Shizuku-based controller features.

<https://github.com/mastercook777/Heimdall-AYN-Thor-Assistant/releases>

1. What Heimdall Is

Heimdall keeps the game on Thor's upper screen while the lower screen becomes a persistent control and reference console. Profiles, Macros, touch controls, maps, guides, static-image Canvases, magnification, recording, and Quick Actions stay within thumb reach.

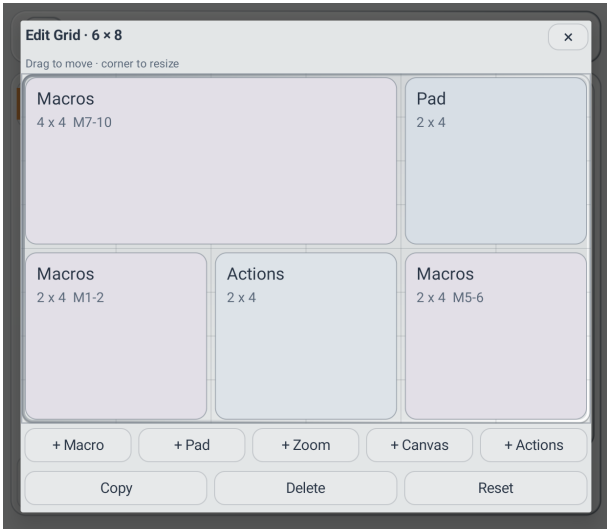
Alpha scope This release targets AYN Thor. Compatibility is not guaranteed for every Thor firmware, controller mode, game, emulator, or other dual-screen device.

Choose only the route you need

What you want to use	Required setup
Profiles, Grid, maps, guides, and Canvas	Install Heimdall only. Developer options and Shizuku are not required.
Upper-screen taps, holds, swipes, or touchpad drag	Enable the Heimdall Accessibility service for Basic Touch.
Controller Macros, Virtual Right Stick, Precision Aim, Enhanced Touch	Install, start, and authorize Shizuku. Developer options are needed only for wireless-debugging startup.
Live magnifier or recording	Accept Android's screen-capture consent when requested.



Play view: Macros, Quick Actions, and touchpad



Edit view: drag modules and resize from visible corners

2. Download, Install, And Migrate

Download and install the APK

- 1 Open the [Heimdall Releases](#) page.
- 2 Open the newest immutable, versioned Alpha release.
- 3 Download the file named like heimdall-vX.Y.Z-alpha.N.apk. The same release also includes SHA256SUMS.txt.
- 4 Open the APK with Thor's file manager.
- 5 If Android asks, allow that file manager to install unknown apps, then return to the installer.
- 6 Finish installation and open Heimdall.

Safe download Install only versioned releases from this repository. Do not trust re-signed APKs or files from unknown sources.

Move from an older test build

The public Alpha package is com.mastercook777.heimdall. Older test packages and new Debug builds may be separate Apps and do not share data with the public package.

- 1 Open Profile management in the old build and export all Profiles.
- 2 Keep the JSON somewhere easy to find.
- 3 Install and open the public Alpha, then re-authorize the services you use.
- 4 Import the Profile JSON and confirm the important Profiles.
- 5 Do not delete the old build or backup until migration is confirmed.

Current export boundary Profile export is configuration JSON. It does not include custom Profile icons, user-imported Macro icons, or Canvas image files; select or import those assets again after migration.

3. Create Your First Profile

- 1 Open Settings > Profile management.
- 2 Create a Profile for one game or emulator.
- 3 Start the target App on the upper screen, then bind it from the recent-App selector.
- 4 Choose a Grid preset. Balanced is the easiest starting point.
- 5 Add only the modules you want now.
- 6 Tap Save at the bottom of Settings, then return to the game and verify Profile selection.

Save boundary Grid and several Profile settings use preview or draft state. Seeing the preview does not mean it has been stored; save before leaving.

4. Basic Touch And Shizuku

Enable Basic Touch

Basic Touch uses Android Accessibility to send compatible taps, holds, swipes, and touchpad drags to the upper screen. It does not require Shizuku or Developer options.

- 1 Open Settings > Connection in Heimdall.
- 2 Find Basic Touch and tap its enable action.
- 3 In Android Accessibility settings, find Heimdall, read the warning, and enable the service.
- 4 Return to Heimdall and confirm Basic Touch is available.
- 5 Test one simple tap Macro or touchpad movement.

Thor mapper coexistence Basic Accessibility touch and Thor's built-in mapper use separate touch streams and may cancel each other. Use Enhanced Touch for coexistence only when Heimdall reports it as available.

Never enabled Developer options?

You need these steps only for Shizuku-based controller features. Menu names can vary by firmware; use Android Settings search for Build number, Developer options, or Wireless debugging when needed.

- 1 Open Thor's Android Settings.
- 2 Open About device, About handheld, or a similarly named page.
- 3 Find Build number. Some firmware places it inside software or version details.
- 4 Tap Build number seven times.
- 5 Enter the screen-lock PIN if asked. Android should confirm that Developer options are enabled.
- 6 Go back to System > Developer options, or search Settings for Developer options.
- 7 Enable USB debugging.
- 8 Open Wireless debugging and turn it on.

Usually one-time You do not need to tap Build number after Developer options are enabled. After a Thor reboot, check Shizuku service state instead.

Pair and start Shizuku

- 1 Install Shizuku from the [official Shizuku Releases page](#).
- 2 Choose Start via wireless debugging, then begin pairing.
- 3 Return to Android's Wireless debugging page and choose Pair device with pairing code.
- 4 Enter the displayed code in Shizuku's notification or pairing screen.
- 5 Return to Shizuku, tap Start, and wait for it to report that Shizuku is running.
- 6 Open Heimdall > Settings > Connection > Controller Enhancement.
- 7 Tap authorize/connect and allow Heimdall in the Shizuku permission dialog.
- 8 Return to Heimdall and confirm Controller Enhancement is available.

After reboot Check Shizuku before using controller features. If the service stopped, start it again; pairing normally does not need to be repeated.

If Shizuku stays on "Searching for pairing service," allow background operation and notifications, keep local networking and wireless debugging available, and try turning wireless debugging off and on once. See the [official Shizuku guide](#) for current advice.

5. Grid, Macros, And Touch Controls



Quick Actions and touch control



Magnifier, Canvas, and Macros

Edit the Grid

Use the visual Grid editor to add, drag, resize, or remove modules. Modules snap to the 6 x 8 Grid. Preview first, then save the Grid and Profile explicitly.

Create a touch Macro

- 1 Add a Macro module to the Grid.
- 2 Hold a Macro button for about 1.8 seconds to open the structured editor.
- 3 Add tap, hold, swipe, or wait steps.
- 4 When capturing coordinates, act directly on the upper-screen target.
- 5 Save the Macro, return to Main, and tap the button to run it.

Physical-controller Macros

Controller recording and replay require Shizuku and an available Controller Enhancement route. Thor's built-in mapper may hide already-mapped buttons. If this happens, begin recording, temporarily return to Android Home to press the button, then return to Heimdall to review and save the sequence.

Mode boundary Enhanced Touch does not replay Macros containing physical-controller steps. Heimdall asks you to switch modes instead of silently using a different route.

Touchpad, Virtual Right Stick, And Precision Aim

- Touch drag: converts lower-screen finger movement into an upper-screen drag.
- Virtual Right Stick: emits Thor right-stick input through Controller Enhancement and should return to neutral immediately after release.
- Precision Aim: is for small camera corrections. Start with low sensitivity and keep the physical stick for large turns.

6. Magnifier, Reference Tools, And Capture

Live upper-screen magnifier

- 1 Add the magnifier module to the Grid.
- 2 Start it and accept Android's screen-capture consent.
- 3 Select the upper-screen region.
- 4 Return to the lower screen, adjust frame rate and zoom, then save the Profile.
- 5 Hold the Main-page magnifier to reselect the region.

Occupancy boundary One live magnifier is supported per Profile. Magnification and recording cannot run at the same time.

Maps, guides, and Canvas

- Maps and guides belong to the current Profile and may reference local images, PDFs, text, or an Interactive Map URL.
- Canvas displays static JPG, PNG, or WebP references. A Profile can contain multiple independent Canvases.
- Double-tap a Canvas for full-screen viewing and hold it to change its image or composition. Full-screen browsing does not overwrite the saved Grid composition.

Screenshot and recording

Quick Actions can capture the upper screen. Recording uses Android MediaProjection and Audio Playback Capture rather than microphone input. The upper App may still forbid its game audio from being captured.

7. Backup, Limitations, And Feedback

Back up before uninstalling

After completing an important Grid or Macro setup, export the current Profile or all Profiles and keep a copy outside Heimdall. Recovery snapshots do not replace a manual export. Android platform backup is disabled for this Alpha; uninstalling removes local App data.

Known Alpha limitations

- Mapping-compatible Shizuku Touch is verified only on tested Thor firmware, game, and mapper combinations.
- Basic Accessibility touch and Thor's built-in mapper may cancel each other during simultaneous operation.
- Thor's mapper may hide mapped controller buttons from the Macro recorder.
- Only one live magnifier is supported per Profile, and magnification cannot share MediaProjection with recording.
- After a lower-screen Tab round trip, the frozen magnifier's Stop marker can occasionally be missing; stopping can also leave the final retained frame visible. This does not mean the frame is still live.
- Profile JSON export does not bundle custom icons or Canvas image files.

Common problems

Problem	First response
Shizuku unavailable	Confirm the service is running and Heimdall is authorized; check again after a Thor reboot.
Touchpad does nothing	Check the current Profile mode, Accessibility or Shizuku state, and that the game is on the upper screen.
Mapper and touch interrupt each other	Basic Touch may conflict with the system mapper; use available Enhanced Touch when coexistence is required.
Mapped button is not recorded	Begin recording, temporarily return to Android Home, and record the button there.
Game slows after a system prompt	Return to the game and touch the upper screen once to restore focus.
Magnifier does not start	Stop recording first; both features cannot own screen capture together.
Recording has no game audio	The upper App may forbid Audio Playback Capture.
Settings were not retained	Tap Save before leaving the Grid editor or Settings.

Send a useful bug report

Use [GitHub Issues](#) and include the Heimdall version, Thor firmware, native/Xbox controller mode, game or emulator, Basic Touch/Shizuku route, whether Thor mapping is enabled, exact reproduction steps, frequency, and screenshots or recordings when possible.

[Official links](#) [Heimdall Releases](#) · [GitHub Issues](#) · [Shizuku setup guide](#)